

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech Branch: Electronics & Telecom/Electronics & Comm/Electronics
Subject Code & Name: BTETPE504E/BTEXPE504E Mixed Signal Design

Semester: V

Max Marks: 60

Date: 17/8/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain sampling theory in detail.	CO1	6
B) What is signal processing? Explain Analog and discrete-time signal processing.	CO2	6
C) Write short note on analog discrete-time filters.	CO1	6
Q.2 Solve Any Two of the following.		12
A) What is switched capacitor filter? Explain in detail.	CO1	6
B) What are The Limitations of Switched-Capacitor Filters?	CO3	6
C) Write short note on applications of switched capacitor filters.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Explain Dual slope ADC in detail.	CO2	6
B) Write short note on Successive approximation ADC	CO2	6
C) What is flash ADC. Explain in detail.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain Voltage-mode signaling.	CO1	6
B) Write short note on current- mode signaling,	CO3	6
C) What is mixed-signal layout? Explain in detail.	CO1	6
Q. 5 Solve Any Two of the following.		12
A) What is PLL and how it works?	CO3	6
B) What is the working principle of frequency synthesizer?	CO2	6
C) Write short note on DLL.	CO3	6